

Circles

Complete the square for the centre and radius.
Then find one right angle: tangent, chord bisector
or semicircle.



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THE TOPIC ON ONE MAP

Circles

01 — Centre and radius

complete the square

from a diameter

$$r^2 > 0$$

02 — Chords and diameters

perpendicular bisector

angle in a semicircle

03 — Tangents

radius $\perp$ tangent

$$b^2 - 4ac = 0$$

04 — Two circles, nearest point

$$\text{touch: } d = r_1 \pm r_2$$

nearest point on the line of centres

THE CIRCLE EQUATION

$$(x - a)^2 + (y - b)^2 = r^2 \quad \text{centre } (a, b), \text{ radius } r$$

$$x^2 + y^2 + 2fx + 2gy + c = 0 \Rightarrow \text{centre } (-f, -g), \quad r^2 = f^2 + g^2 - c$$

01 Centre and Radius: Complete the Square

Group the x terms and the y terms and complete the square on each. The numbers inside the brackets change sign for the centre; the number on the right is r^2 — so it must be positive.

WHEN YOU SEE → YOU DO

“find the centre and the exact radius”

→ complete the square in x and in y

“state the possible range of values for k ”

→ $r^2 > 0$ gives an inequality in k

“ $y = k$ is a tangent”

→ $k = b \pm r$ (top and bottom of the circle)

“the x -axis is a tangent”

→ $r = |b|$

“a chord of length 4”

→ half-chord, radius and distance to centre: Pythagoras

WORKED MODEL • JANUARY 2020

$x^2 + y^2 + 6x - 4y - 14 = 0$. Centre and exact radius.

1 $(x + 3)^2 - 9 + (y - 2)^2 - 4 - 14 = 0$

complete both squares

2 $(x + 3)^2 + (y - 2)^2 = 27$

ANSWER centre $(-3, 2)$, $r = \sqrt{27} = 3\sqrt{3}$

IDEA 01

Practise: Centre and Radius: Complete the Square

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Complete the square twice

P2 · TOPIC 3 3 marks

Find the centre and the radius of the circle $x^2 + y^2 - 8x + 2y - 8 = 0$.

Q2 **DR ESLAM** From a diameter

P2 · TOPIC 3 3 marks

The points $(-1, 3)$ and $(5, 11)$ are the ends of a diameter. Find an equation of the circle.

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P2 · TOPIC 3**3 marks**

1 Write down the equations of the circles with the following centres and radii

(i) Centre: $(0, 0)$ Radius: $r = 4$,

(ii) Centre: $(3, -4)$ Radius: $r = 2$,

(iii) Centre: $(-5, 0)$ Radius: $r = 5$.

(3 marks)**Q4****PRACTICE**

Practice · Medium

P2 · TOPIC 3**4 marks**

1 A circle has centre $(6, -5)$ and goes through the point $(1, 7)$. Find the equation of the circle.

(4 marks)

Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 3**4 marks**

2 Find the centre and radius of the circle with equation $x^2 + y^2 + x - 3y + 2 = 0$.

(4 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA12 · January 2020 · Q6

P2 · TOPIC 3 8 marks

6. The circle C has equation

$$x^2 + y^2 + 6x - 4y - 14 = 0$$

(a) Find

(i) the coordinates of the centre of C ,(ii) the exact radius of C .

(3)

The line with equation $y = k$, where k is a constant, is a tangent to C .(b) Find the possible values of k .

(2)

The line with equation $y = p$, where p is a negative constant, is a chord of C .

Given that the length of this chord is 4 units,

(c) find the value of p .

(3)

(Total 8 marks)

Q7 **PAST PAPER** Pearson IAL WMA12 · June 2021 · Q6**P2 · TOPIC 3** **7 marks**

6. A circle has equation

$$x^2 - 6x + y^2 + 8y + k = 0$$

where k is a positive constant.

Given that the x -axis is a tangent to this circle,

- (a) find the value of k .

(3)

The circle meets the coordinate axes at the points R , S and T .

- (b) Find the exact area of the triangle RST .

(4)

(Total 7 marks)

Q8 PAST PAPER Pearson IAL WMA12 · October 2021 · Q6

P2 · TOPIC 3 8 marks

6. (i) The circle C_1 has equation

$$x^2 + y^2 + 10x - 12y = k \quad \text{where } k \text{ is a constant}$$

- (a) Find the coordinates of the centre of C_1 (2)

- (b) State the possible range in values for k . (2)

- (ii) The point $P(p, 0)$, the point $Q(-2, 10)$ and the point $R(8, -14)$ lie on a different circle, C_2

Given that

- p is a positive constant
- QR is a diameter of C_2

find the exact value of p . (4)

(Total 8 marks)

Q9 PAST PAPER Pearson IAL WMA12 · January 2024 · Q3

P2 · TOPIC 3 6 marks

3. The circle C

- has centre $A(3, 5)$
- passes through the point $B(8, -7)$

(a) Find an equation for C .

(3)

The points M and N lie on C such that MN is a chord of C .Given that MN

- lies above the x -axis
- is parallel to the x -axis
- has length $4\sqrt{22}$

(b) find an equation for the line passing through points M and N .

(3)

(Total for Question 3 is 6 marks)

02 Chords, Diameters and Right Angles

Two circle facts turn into straight-line work. The perpendicular bisector of **any** chord passes through the centre. The angle in a semicircle is 90° — so if PQR is a right angle, PR is a diameter.

WHEN YOU SEE → YOU DO

“show that angle $PQR = 90^\circ$ ”

→ $m_{PQ} \times m_{QR} = -1$

“hence find the centre”

→ the right angle makes PR a diameter:
centre = midpoint

“the line through M (midpoint of AB) and P ”

→ perpendicular bisector: gradient $-\frac{1}{m_{AB}}$

“ $P(7, k)$ lies on l ”

→ substitute into the bisector

three points on a circle

→ two perpendicular bisectors meet at the centre

ZONE DIAMETER

ends A, B

centre = midpoint

ZONE CHORD

AB

bisector passes through the centre

ZONE SEMICIRCLE

angle at Q

$90^\circ \Leftrightarrow PR$ is a diameter

WORKED MODEL • JANUARY 2025

$A(-1, 2)$ and $B(11, 6)$ lie on a circle, centre P . Find the line l through M (midpoint) and P .

1 $M(5, 4), m_{AB} = \frac{4}{12} = \frac{1}{3}$

midpoint, gradient

2 $m_l = -3: y - 4 = -3(x - 5)$

perpendicular bisector

ANSWER $y = -3x + 19$

IDEA 02

Practise: Chords, Diameters and Right Angles

My questions first, then selected practice, then real past papers.

Q10 **DR ESLAM** The perpendicular bisector

P2 · TOPIC 3 4 marks

$A(1, 2)$ and $B(5, 6)$ lie on a circle whose centre is on the x -axis. Find an equation of the circle.

The perpendicular bisector of any chord passes through the centre.

SELECTED PRACTICE

Q11**PRACTICE**

Practice · Medium

P2 · TOPIC 3 7 marks

6 (a) The points $A(3, 5)$, $B(5, 3)$ and $C(9, 7)$ lie on a circle.

Show that triangle ABC is a right-angle triangle.

(2 marks)

(b) Explain why the line segment AC must be the diameter of the circle.

(1 mark)

(c) Hence find the equation of the circle.

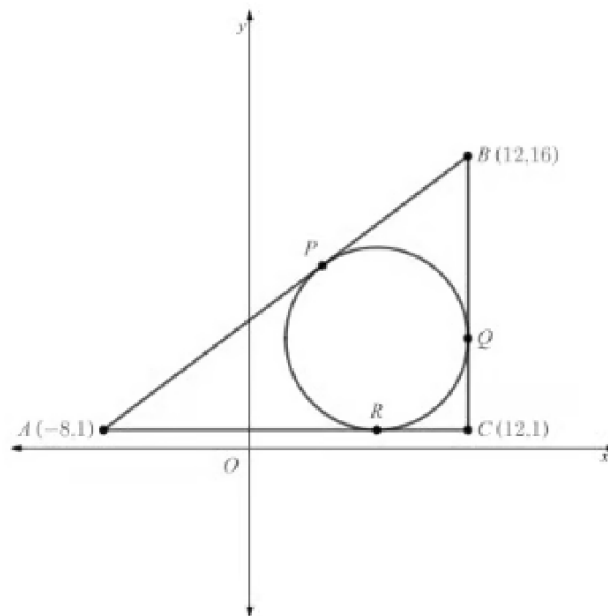
(4 marks)

Q12**PRACTICE**

Practice · Hard

P2 · TOPIC 3 7 marks

- 7 (a)** Triangle ABC has vertices $A(-8, 1)$, $B(12, 16)$ and $C(12, 1)$. A circle with equation $(x - 7)^2 + (y - 6)^2 = 25$ touches Triangle ABC at the three points P , Q and R , as shown in the diagram below:



Write down the coordinates of points R and Q .

(2 marks)

- (b)** Find the coordinates of point P .

(5 marks)

Q13**PRACTICE**

Practice · Very Hard

P2 · TOPIC 3 7 marks

6 The points $A(4, 6)$, $B(7, 2)$ and $C(12, 12)$ lie on a circle.

Find the equation of the circle.

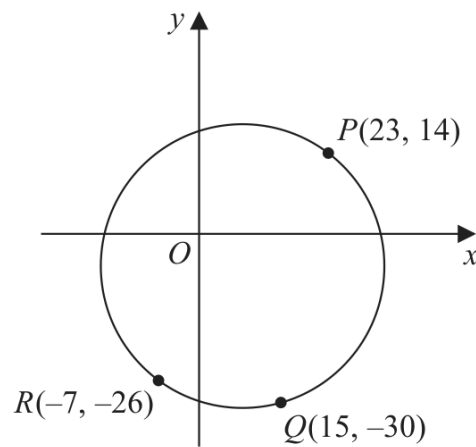
(7 marks)

PAST PAPERS

Q14 PAST PAPER Pearson IAL WMA12 · January 2022 · Q6

P2 · TOPIC 3 8 marks

6.

**Figure 1**

The points $P(23, 14)$, $Q(15, -30)$ and $R(-7, -26)$ lie on the circle C , as shown in Figure 1.

(a) Show that angle $PQR = 90^\circ$ (2)

(b) Hence, or otherwise, find

- (i) the centre of C ,
- (ii) the radius of C . (3)

Given that the point S lies on C such that the distance QS is greatest,

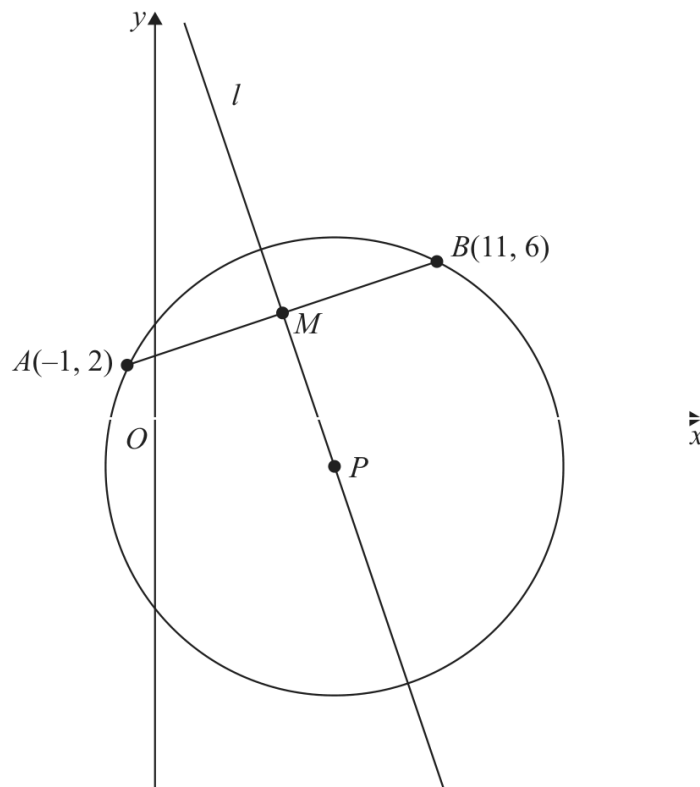
(c) find an equation of the tangent to C at S , giving your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found. (3)

(Total 8 marks)

Q15 PAST PAPER Pearson IAL WMA12 · January 2025 · Q6

P2 · TOPIC 3 8 marks

6.

**Figure 2**

The point $A(-1, 2)$ and the point $B(11, 6)$ both lie on a circle with centre P .

The point M is the midpoint of AB .

Given that the line l passes through M and P , as shown in Figure 2,

- (a) find an equation for l , giving your answer in the form $y = mx + c$, where m and c are constants. (4)

Given that P has coordinates $(7, k)$, where k is a constant,

- (b) find the value of k , (1)
- (c) find an equation for the circle. (3)

(Total for Question 6 is 8 marks)

03 Tangents and Lines Meeting a Circle

A tangent is perpendicular to the radius at the point of contact. When the point is unknown, substitute the line into the circle: a tangent gives one repeated root, so $b^2 - 4ac = 0$.

WHEN YOU SEE → YOU DO

“the tangent to C at P”

→ radius gradient, then $-\frac{1}{m}$, then $y - y_1 = m(x - x_1)$

“ $y = mx + 1$ is a tangent to C”

→ substitute, then $b^2 - 4ac = 0$

“find the points where the line meets the circle”

→ substitute the line, solve the quadratic

“the tangent ... in the form $ax + by + c = 0$ ”

→ integers: clear the fractions

“the line through the centre and the point of contact”

→ it is the normal: perpendicular to the tangent

Centre and radius

Gradient of the radius

Negative reciprocal

Line through the point

WORKED MODEL • JUNE 2023

Centre (2, 5), $P(8, -3)$ on the circle. Tangent at P.

1 radius gradient $\frac{-3-5}{8-2} = -\frac{4}{3}$

centre to P

2 tangent gradient $\frac{3}{4}$: $y + 3 = \frac{3}{4}(x - 8)$

perpendicular

ANSWER $3x - 4y - 36 = 0$

IDEA 03

Practise: Tangents and Lines Meeting a Circle

My questions first, then selected practice, then real past papers.

Q16 DR ESLAM Tangent at a point

P2 · TOPIC 3 3 marks

Find the tangent to $(x - 2)^2 + (y + 1)^2 = 25$ at $(5, 3)$, in the form $ax + by + c = 0$.

Q17 DR ESLAM Tangent by the discriminant

P2 · TOPIC 3 4 marks

Find the values of k for which $y = x + k$ is a tangent to $x^2 + y^2 = 8$.

SELECTED PRACTICE

Q18 **PRACTICE** Practice · Easy

P2 · TOPIC 3 2 marks

7 On a sketch show how a circle and a line can either have 0, 1 or 2 intersections.

(2 marks)**Q19** **PRACTICE** Practice · Hard

P2 · TOPIC 3 4 marks

3 The line $y + 2x = 11$ meets the circle with equation $x^2 + y^2 + 6x - 14y = -38$.

(i) Show that the line and circle meet at one point only.

(ii) Find the coordinates of the point of intersection.

(4 marks)

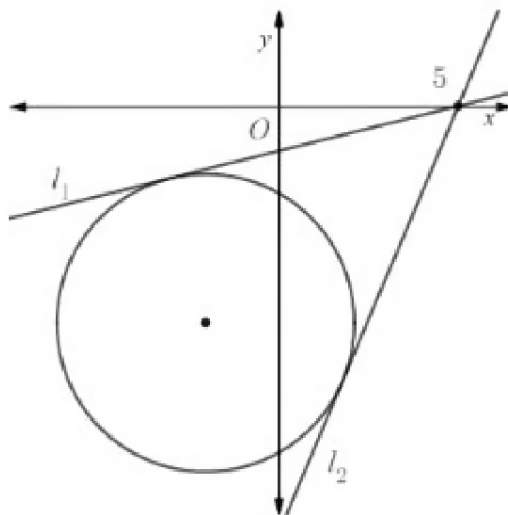
Q20**PRACTICE**

Practice · Very Hard

P2 · TOPIC 3 8 marks

7 A circle has equation $x^2 + y^2 + 4x + 12y = -23$.

The lines l_1 and l_2 are both tangents to the circle, and they intersect at the point $(5, 0)$.



Find the equations of l_1 and l_2 , giving your answers in the form $y = mx + c$.

(8 marks)

PAST PAPERS

Q21 PAST PAPER Pearson IAL WMA12 · October 2019 · Q6

P2 · TOPIC 3 7 marks

6.

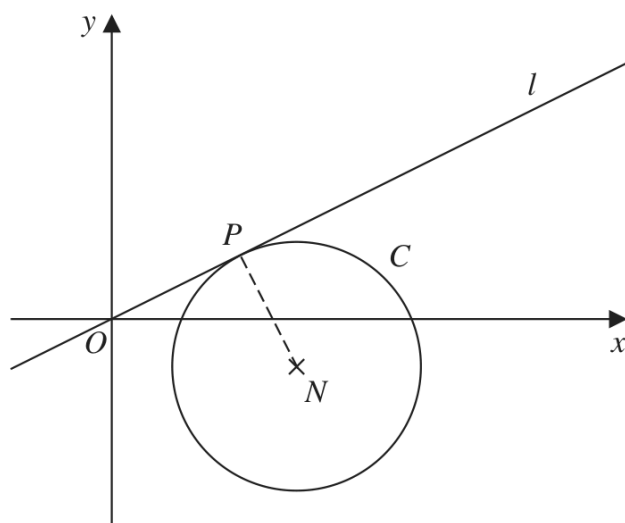
**Figure 1**

Figure 1 shows a sketch of a circle C with centre $N(4, -1)$.

The line l with equation $y = \frac{1}{2}x$ is a tangent to C at the point P .

Find

(a) the equation of line PN in the form $y = mx + c$, where m and c are constants, (2)

(b) the equation of C . (5)

(Total 7 marks)

Q22 PAST PAPER Pearson IAL WMA12 · January 2021 · Q9

P2 · TOPIC 3 10 marks

9. A circle C has equation

$$(x - k)^2 + (y - 2k)^2 = k + 7$$

where k is a positive constant.

(a) Write down, in terms of k ,

(i) the coordinates of the centre of C ,

(ii) the radius of C .

(2)

Given that the point $P(2, 3)$ lies on C

(b) (i) show that $5k^2 - 17k + 6 = 0$

(ii) hence find the possible values of k .

(3)

The tangent to the circle at P intersects the x -axis at point T .

Given that $k < 2$

(c) calculate the exact area of triangle OPT .

(5)

(Total 10 marks)

Q23 PAST PAPER Pearson IAL WMA12 · June 2022 · Q10

P2 · TOPIC 3 12 marks

10. The circle C has centre $X(3, 5)$ and radius r

The line l has equation $y = 2x + k$, where k is a constant.

(a) Show that l and C intersect when

$$5x^2 + (4k - 26)x + k^2 - 10k + 34 - r^2 = 0 \quad (3)$$

Given that l is a tangent to C ,

(b) show that $5r^2 = (k + p)^2$, where p is a constant to be found. (3)

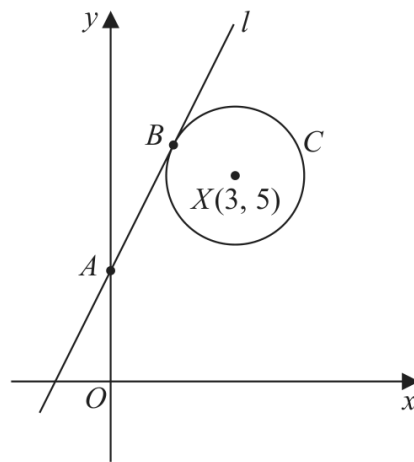


Figure 2

The line l

- cuts the y -axis at the point A
- touches the circle C at the point B

as shown in Figure 2.

Given that $AB = 2r$

(c) find the value of k (6)

9. In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

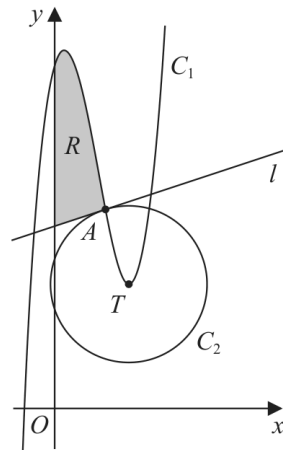


Figure 3

Figure 3 shows

- the curve C_1 with equation $y = x^3 - 5x^2 + 3x + 14$
- the circle C_2 with centre T

The point T is the minimum turning point of C_1

Using Figure 3 and calculus,

(a) find the coordinates of T

(3)

The curve C_1 intersects the circle C_2 at the point A with x coordinate 2

(b) Find an equation of the circle C_2

(3)

The line l shown in Figure 3, is the tangent to circle C_2 at A

(c) Show that an equation of l is

$$y = \frac{1}{3}x + \frac{22}{3}$$

(3)

The region R , shown shaded in Figure 3, is bounded by C_1 , l and the y -axis.

(d) Find the exact area of R .

(3)

(Total for Question 9 is 12 marks)

Q25 **PAST PAPER** Pearson IAL WMA12 · January 2023 · Q6**P2 · TOPIC 3** **10 marks**

6. The circle C has equation

$$x^2 + y^2 + 8x - 4y = 0$$

(a) Find

- (i) the coordinates of the centre of C ,
- (ii) the exact radius of C .

(3)

The point P lies on C .

Given that the tangent to C at P has equation $x + 2y + 10 = 0$

(b) find the coordinates of P

(4)

(c) Find the equation of the normal to C at P , giving your answer in the form $y = mx + c$ where m and c are integers to be found.

(3)

(Total for Question 6 is 10 marks)

Q26 **PAST PAPER** Pearson IAL WMA12 · June 2023 · Q3**P2 · TOPIC 3** **7 marks**

3. A circle C has centre $(2, 5)$

Given that the point $P(8, -3)$ lies on C

(a) (i) find the radius of C

(ii) find an equation for C

(3)

(b) Find the equation of the tangent to C at P giving your answer in the form $ax + by + c = 0$ where a , b and c are integers to be found.

(4)

(Total for Question 3 is 7 marks)

7.

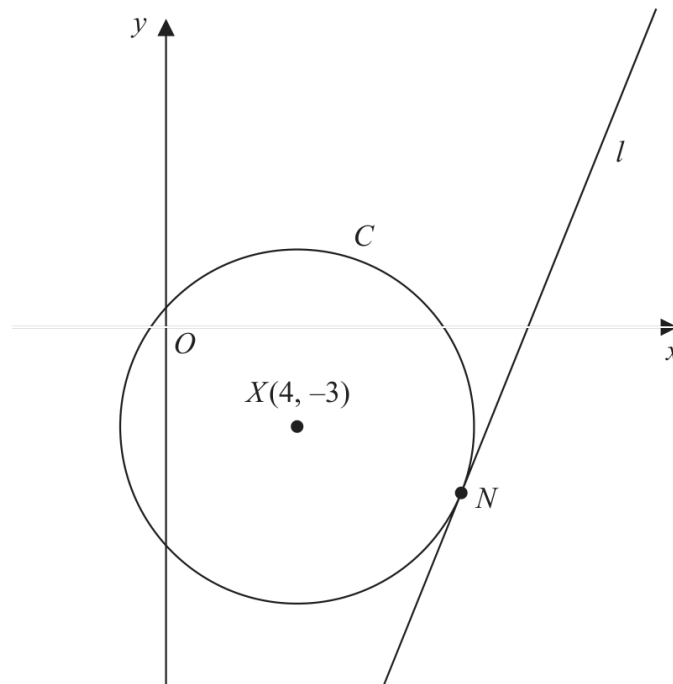


Figure 2

Figure 2 shows a sketch of

- the circle C with centre $X(4, -3)$
- the line l with equation $y = \frac{5}{2}x - \frac{55}{2}$

Given that l is the tangent to C at the point N ,

(a) show that an equation for the straight line passing through X and N is

$$2x + 5y + 7 = 0 \quad (3)$$

(b) Hence find

- the coordinates of N ,
- an equation for C .

(5)

(Total for Question 7 is 8 marks)

Q28 **PAST PAPER** Pearson IAL WMA12 · October 2024 · Q10**P2 · TOPIC 3** **10 marks**

10. The circle C has equation

$$x^2 + y^2 + 4x - 30y + 209 = 0$$

(a) Find

- (i) the coordinates of the centre of C ,
- (ii) the exact value of the radius of C .

(3)

The line L has equation $y = mx + 1$, where m is a constant.

Given that L is the tangent to C at the point P ,

(b) show that

$$2m^2 - 7m - 22 = 0$$

(3)

(c) Hence find the possible pairs of coordinates of P .

(4)

(Total for Question 10 is 10 marks)

04 Two Circles and the Nearest Point

Everything happens on the line joining the centres. Two circles touch when the distance between the centres is $r_1 + r_2$ (outside) or $|r_1 - r_2|$ (inside). The nearest point of a circle to a point O is on the line from O to the centre, at distance $d - r$.

WHEN YOU SEE → YOU DO

“the circles touch”

$$\rightarrow d = r_1 + r_2 \text{ or } d = |r_1 - r_2|$$

“neither touch nor intersect”

$$\rightarrow d > r_1 + r_2 \text{ (or } d < |r_1 - r_2| \text{)}$$

“the point P on the circle nearest the origin”

$$\rightarrow OP = d - r \text{ along the line to the centre}$$

“nearest the x -axis”

$$\rightarrow \text{straight below the centre: } (a, b - r)$$

WORKED MODEL • JUNE 2024

C_1 : centre $(-4, 5)$, $r = \sqrt{70}$. C_2 : centre $(5, -8)$, $r = \sqrt{52}$. Do they meet?

1 $d = \sqrt{81 + 169} = \sqrt{250} \approx 15.8$

distance between centres

2 $r_1 + r_2 \approx 8.37 + 7.21 = 15.6$

sum of radii

ANSWER $d > r_1 + r_2$: they neither touch nor intersect

× **TRAP** Comparing d^2 with $r_1^2 + r_2^2$. Compare d with $r_1 + r_2$.

IDEA 04

Practise: Two Circles and the Nearest Point

My questions first, then selected practice, then real past papers.

Q29 DR ESLAM Do they touch?

P2 · TOPIC 3 3 marks

Show that the circles $x^2 + y^2 = 4$ and $(x - 6)^2 + (y - 8)^2 = 25$ do not meet, and find the shortest distance between them.

SELECTED PRACTICE**Q30** PRACTICE Practice · Easy

P2 · TOPIC 3 3 marks

6 Determine if the circles with equations

$$(x + 4)^2 + y^2 = 9 \text{ and } (x - 2)^2 + y^2 = 9$$

intersect once, twice or not at all. Fully explain your answer.

(3 marks)

Q31 PRACTICE Practice · Medium

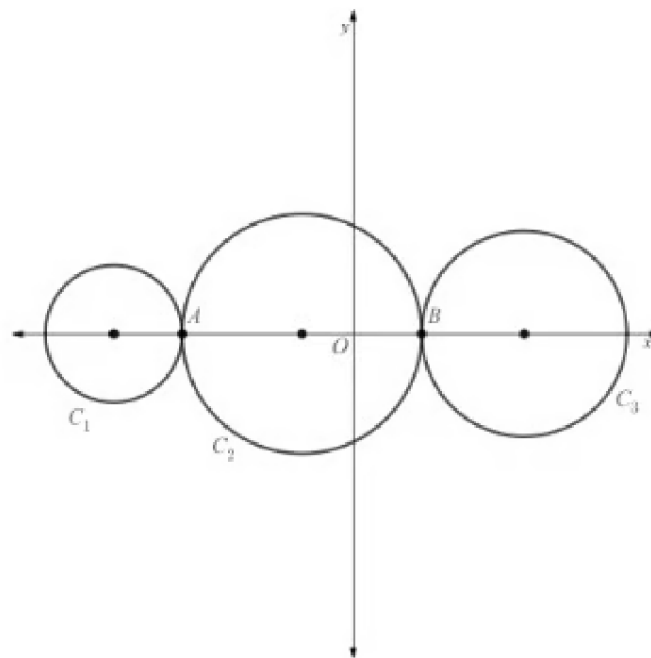
P2 · TOPIC 3 6 marks

7 Circles C_1 , C_2 and C_3 all have their centres on the x -axis.

Circle C_1 has equation $(x + 7)^2 + y^2 = 4$.

Circle C_3 has equation $x^2 + y^2 - 10x + 16 = 0$.

Circles C_1 and C_2 touch at point A , and circles C_2 and C_3 touch at point B .



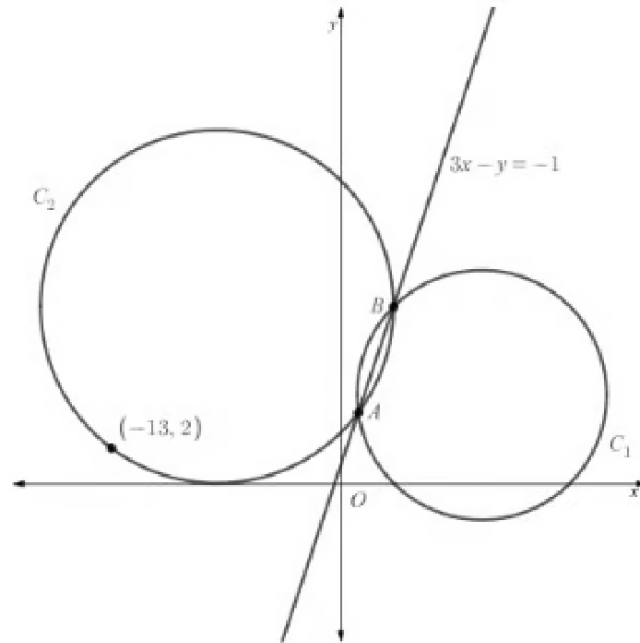
Find the coordinates of the centre of circle C_2 .

(6 marks)

Q32 **PRACTICE** Practice · Very Hard

P2 · TOPIC 3 11 marks

- 8 The diagram below shows circles C_1 and C_2 which intersect at the two points A and B . Circle C_1 has equation $x^2 + y^2 - 16x - 10y + 39 = 0$, and points A and B lie along the line with equation $3x - y = -1$. Circle C_2 also passes through the point $(-13, 2)$.



Find an equation of circle C_2 .

(11 marks)

PAST PAPERS

Q33 PAST PAPER Pearson IAL WMA12 · June 2024 · Q7

P2 · TOPIC 3 6 marks

7. The circle C_1 has equation

$$x^2 + y^2 + 8x - 10y = 29$$

(a) (i) Find the coordinates of the centre of C_1

(ii) Find the exact value of the radius of C_1

(3)

In part (b) you must show detailed reasoning.

The circle C_2 has equation

$$(x - 5)^2 + (y + 8)^2 = 52$$

(b) Prove that the circles C_1 and C_2 neither touch nor intersect.

(3)

(Total for Question 7 is 6 marks)

Q34 PAST PAPER Pearson IAL WMA12 · June (R paper) 2024 · Q3

P2 · TOPIC 3 5 marks

3. A circle has equation

$$x^2 + y^2 + 8x - 14y - 79 = 0$$

(a) Find

- (i) the coordinates of the centre of the circle,
- (ii) the radius of the circle.

(3)

Given that P is the point on the circle that is nearest the origin O ,

- (b) find the exact length of OP

(2)

(Total for Question 3 is 5 marks)

Q35 **PAST PAPER** Pearson IAL WMA12 · June 2025 · Q2**P2 · TOPIC 3** **7 marks**

2: The line joining the points $(-2, 5)$ and $(4, 15)$ is the diameter of a circle C .

(a) Find an equation for C .

(5)

(b) Hence find the exact coordinates of the point on C that is nearest the x -axis.

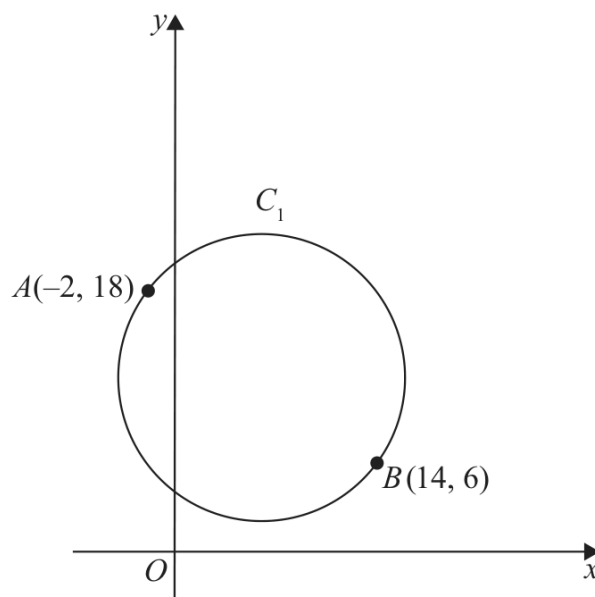
(2)

(Total for Question 2 is 7 marks)

Q36 PAST PAPER Pearson IAL WMA12 · October 2025 · Q6

P2 · TOPIC 3 9 marks

6.

**Figure 2**

The points $A(-2, 18)$ and $B(14, 6)$ lie on a circle C_1 as shown in Figure 2.

Given that AB is a diameter of the circle C_1

- (a) find an equation for C_1 making your method clear. (5)

A circle C_2 has its centre at the origin.

Given that circles C_1 and C_2 touch,

- (b) find possible equations for C_2 (4)

(Total for Question 6 is 9 marks)

Q37 PAST PAPER Pearson IAL WMA12 · January 2026 · Q5

P2 · TOPIC 3 6 marks

5. The circle C_1 has equation

$$x^2 + y^2 - 6x + 5y - 41 = 0$$

(a) Find

- (i) the coordinates of the centre of C_1
- (ii) the radius of C_1

(3)

The circle C_2 has

- centre $(-k, 0)$ where k is a positive constant
- radius 5

Given that circles C_1 and C_2 touch

(b) find the exact value of k .

(3)

(Total for Question 5 is 6 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------------|---|
| 01 | Centre, radius | Complete the square; $r^2 > 0$. |
| 02 | Chords, diameters | Bisector through the centre; semicircle 90° . |
| 03 | Tangents | Radius $\perp$ tangent; unknown point $\rightarrow b^2 - 4ac = 0$. |
| 04 | Two circles | $d = r_1 \pm r_2$; nearest point at $d - r$. |

BOTTOM LINE Centre and radius first · then one right angle: tangent, semicircle or chord bisector.

Answers

Circles · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · CENTRE AND RADIUS: COMPLETE THE SQUARE

- Q1** Dr Eslam · Complete the square twice $(x - 4)^2 + (y + 1)^2 = 25$: centre $(4, -1)$, radius 5
- Q2** Dr Eslam · From a diameter centre $(2, 7)$, $r = 5$: $(x - 2)^2 + (y - 7)^2 = 25$
- Q3** Practice · Easy (i) $x^2 + y^2 = 16$ (ii) $(x - 3)^2 + (y + 4)^2 = 4$ (iii) $(x + 5)^2 + y^2 = 25$
- Q4** Practice · Medium $r^2 = 5^2 + 12^2 = 169$: $(x - 6)^2 + (y + 5)^2 = 169$
- Q5** Practice · Very Hard $(x + \frac{1}{2})^2 + (y - \frac{3}{2})^2 = \frac{1}{2}$: centre $(-\frac{1}{2}, \frac{3}{2})$, radius $\frac{\sqrt{2}}{2}$
- Q6** Pearson IAL WMA12 · January 2020 · Q6 (a)(i) $(-3, 2)$. (a)(ii) $3\sqrt{3}$. (b) $k = 2 \pm 3\sqrt{3}$. (c) $p = 2 - \sqrt{23}$
- Q7** Pearson IAL WMA12 · June 2021 · Q6 (a) $k = 9$. (b) $3\sqrt{7}$
- Q8** Pearson IAL WMA12 · October 2021 · Q6 (i)(a) $(-5, 6)$. (i)(b) $k > -61$. (ii) $p = 3 + \sqrt{165}$
- Q9** Pearson IAL WMA12 · January 2024 · Q3 (a) $(x - 3)^2 + (y - 5)^2 = 169$. (b) $y = 14$

IDEA 02 · CHORDS, DIAMETERS AND RIGHT ANGLES

- Q10** Dr Eslam · The perpendicular bisector
 bisector $y = 7 - x$; centre $(7, 0)$: $(x - 7)^2 + y^2 = 40$
- Q11** Practice · Medium
 (a) $m_{AB} = -1$, $m_{BC} = 1$: perpendicular, right angle at B (b) the angle in a semicircle is 90°
 (c) $(x - 6)^2 + (y - 6)^2 = 10$
- Q12** Practice · Hard
 (a) $R(7, 1)$, $Q(12, 6)$ (b) $P(4, 10)$
- Q13** Practice · Very Hard
 $x^2 + y^2 - 19x - 14y + 108 = 0$, i.e. $(x - \frac{19}{2})^2 + (y - 7)^2 = \frac{125}{4}$
- Q14** Pearson IAL WMA12 · January 2022 · Q6
 (a) Proven: $\angle PQR = 90^\circ$.
 (b)(i) $(8, -6)$. (b)(ii) 25. (c) $7x - 24y + 425 = 0$
- Q15** Pearson IAL WMA12 · January 2025 · Q6
 (a) $y = -3x + 19$. (b) $k = -2$. (c) $(x - 7)^2 + (y + 2)^2 = 80$

IDEA 03 · TANGENTS AND LINES MEETING A CIRCLE

- Q16** Dr Eslam · Tangent at a point
radius gradient $\frac{4}{3}$, tangent $-\frac{3}{4}$: $3x+4y-27=0$
- Q17** Dr Eslam · Tangent by the discriminant
 $2x^2 + 2kx + k^2 - 8 = 0$; $b^2 - 4ac = 0 \Rightarrow k = \pm 4$
- Q18** Practice · Easy
misses (0), tangent (1), chord (2)
- Q19** Practice · Hard
(i) the quadratic has a repeated root (ii) (1, 9)
 $m = \frac{1}{4}$ or $\frac{19}{8}$: $y = \frac{1}{4}x - \frac{5}{4}$, $y = \frac{19}{8}x - \frac{95}{8}$
- Q20** Practice · Very Hard
(a) $y = -2x + 7$. (b) $(x-4)^2 + (y+1)^2 = \frac{36}{5}$
- Q21** Pearson IAL WMA12 · October 2019 · Q6
(a)(i) $(k, 2k)$. (a)(ii) $\sqrt{k+7}$. (b)(ii) $k = 3$ or $\frac{2}{5}$. (c) $\frac{147}{16}$
- Q22** Pearson IAL WMA12 · January 2021 · Q9
(a) Proven. (b) $5r^2 = (k+1)^2$, so $p = 1$.
(c) $k = \frac{11}{4}$
- Q23** Pearson IAL WMA12 · June 2022 · Q10
(a) $T = (3, 5)$. (b) $(x-3)^2 + (y-5)^2 = 10$. (c) $y = \frac{1}{3}x + \frac{22}{3}$. (d) $\frac{28}{3}$
- Q24** Pearson IAL WMA12 · October 2022 · Q9
(a)(i) $(-4, 2)$. (a)(ii) $2\sqrt{5}$. (b) $P = (-6, -2)$.
(c) $y = 2x + 10$
- Q25** Pearson IAL WMA12 · January 2023 · Q6
(a)(i) 10. (a)(ii) $(x-2)^2 + (y-5)^2 = 100$. (b) $3x - 4y - 36 = 0$
- Q26** Pearson IAL WMA12 · June 2023 · Q3
(a) Proven. (b)(i) $N = (9, -5)$. (b)(ii) $(x-4)^2 + (y+3)^2 = 29$
- Q27** Pearson IAL WMA12 · October 2023 · Q7
(a)(i) $(-2, 15)$. (a)(ii) $2\sqrt{5}$. (b) Proven.
(c) $(-6, 13)$ and $(\frac{12}{5}, \frac{71}{5})$
- Q28** Pearson IAL WMA12 · October 2024 · Q10

IDEA 04 · TWO CIRCLES AND THE NEAREST POINT

- Q29** Dr Eslam · Do they touch? $d = 10 > 2 + 5$; shortest distance $10 - 7 = 3$
- Q30** Practice · Easy centres $(-4, 0)$, $(2, 0)$: distance $6 = 3 + 3$, so they touch once
- Q31** Practice · Medium $A(-5, 0)$, $B(2, 0)$: centre of C_2 is $(-\frac{3}{2}, 0)$
- Q32** Practice · Very Hard $A(1, 4)$, $B(3, 10)$; $C_2: (x + 7)^2 + (y - 10)^2 = 100$
- Q33** Pearson IAL WMA12 · June 2024 · Q7 $(a)(i) (-4, 5)$. $(a)(ii) \sqrt{70}$. (b) The circles neither touch nor intersect
- Q34** Pearson IAL WMA12 · June (R paper) 2024 · Q3 $(a)(i) (-4, 7)$. $(a)(ii) 12$. $(b) 12 - \sqrt{65}$
- Q35** Pearson IAL WMA12 · June 2025 · Q2 $(a) (x - 1)^2 + (y - 10)^2 = 34$. $(b) (1, 10 - \sqrt{34})$
- Q36** Pearson IAL WMA12 · October 2025 · Q6 $(a) (x - 6)^2 + (y - 12)^2 = 100$. $(b) x^2 + y^2 = 280 - 120\sqrt{5}$ or $x^2 + y^2 = 280 + 120\sqrt{5}$
- Q37** Pearson IAL WMA12 · January 2026 · Q5 $(a)(i) (3, -\frac{5}{2})$. $(a)(ii) \frac{15}{2}$. $(b) k = 5\sqrt{6} - 3$